



Laboratory Work 2

Synthesizing Controller for the given cost function

Lab Goal: *To Synthesize a controller for the given cost function for the optimal performance*

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Lab Variant: 28

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1. Introduction

In this laboratory work we have to synthesize a controller for the given cost function on a specific time interval to produce the optimal solution, we have a plant, a cost function and a set of initial and constraint values that we have to work upon, we will use the Hamiltonian equation and the Pontryagin's maximum principle for the solution of the system

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -6x_1 + u \end{cases} \rightarrow \text{Plant}$$

$$J = \int_0^4 u^2(\tau) d\tau \rightarrow \text{Cost function}$$

Where,

$$x_1(0) = x_2(0) = 0$$

$$x_1(4) = 70, x_2(4) = 0$$

2. Hamiltonian Equation

We will now form the Hamiltonian equation for our system,

$$H = \phi_0 u^2 + \phi \dot{x} = -u^2 + \phi_1 \dot{x}_1 + \phi_2 \dot{x}_2$$

So our equation will take the form,

$$H = -u^2 + \phi_1 x_2 + \phi_2 (6x_1 + u)$$

3. Pontryagins Maximum Principle (Euler Lagrange)

Now we will use the Euler lagrange equations and the pontryagin's maximum principle to form the differential equations of our system to later synthesize the control law u ,

$$\begin{cases} \dot{\phi}_i = -\frac{\partial H}{\partial x_i} \\ \frac{\partial H}{\partial u} = 0 \end{cases}$$

By calculating the required partial derivatives

$$\begin{cases} \dot{\phi}_1 = 6\phi_2 \\ \dot{\phi}_2 = -\phi_1 \\ -2u + \phi_2 = 0 \end{cases} \rightarrow \begin{cases} \dot{\phi}_1 = 6\phi_2 \\ \dot{\phi}_2 = -\phi_1 \\ \dot{x}_1 = x_2 \\ \dot{x}_2 = -6x_1 + \frac{\phi_2}{2} \end{cases}$$

Now we will solve these equations using matlab to synthesize the control law (u) that will be used to get the optimal control

As we know that we have calculated u as,

$$u = \frac{\phi_2}{2}$$

But we need to solve the above equations for finding the value of ϕ_2

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syms f1(t) f2(t) x1(t) x2(t)
ode1 = diff(f1) == 6*f2;
ode2 = diff(f2) == f1;
ode3 = diff(x1) == x2;
ode4 = diff(x2) == (-6*x1)+(f2/2);
odes = [ode1; ode2;ode3; ode4];
cond1 = x1(0) == 0;
cond2 = x2(0) == 0;
cond3 = x1(4) == 70;
cond4 = x2(4) == 0;
conds = [cond1; cond2;cond3; cond4];
[f1Sol(t),f2Sol(t),x1Sol(t),x2Sol(t)] = dsolve(odes,conds);
u=f2Sol(t)/2;
disp(u)

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Figure 1. Calculation for control law

Since we have now calculated the control law, we will model the system to run the plant with the controller to achieve the required result,

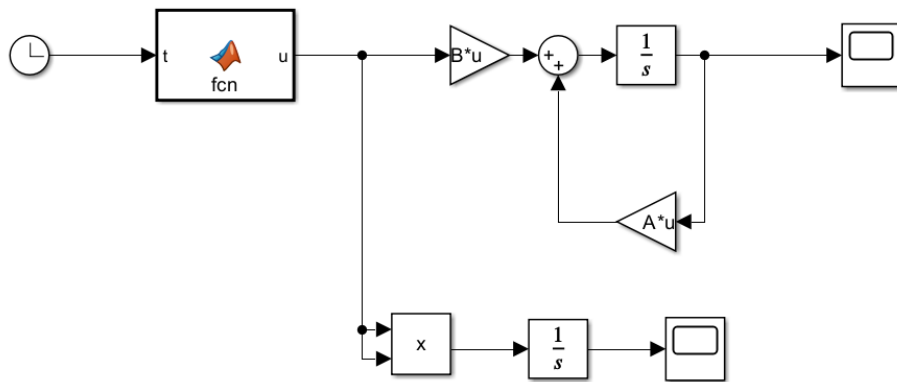


Figure 2. System Model

Here A and B are our state and input matrices which are as per our plant equations,

$$A = \begin{bmatrix} 0 & 1 \\ -6 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

4. Outputs

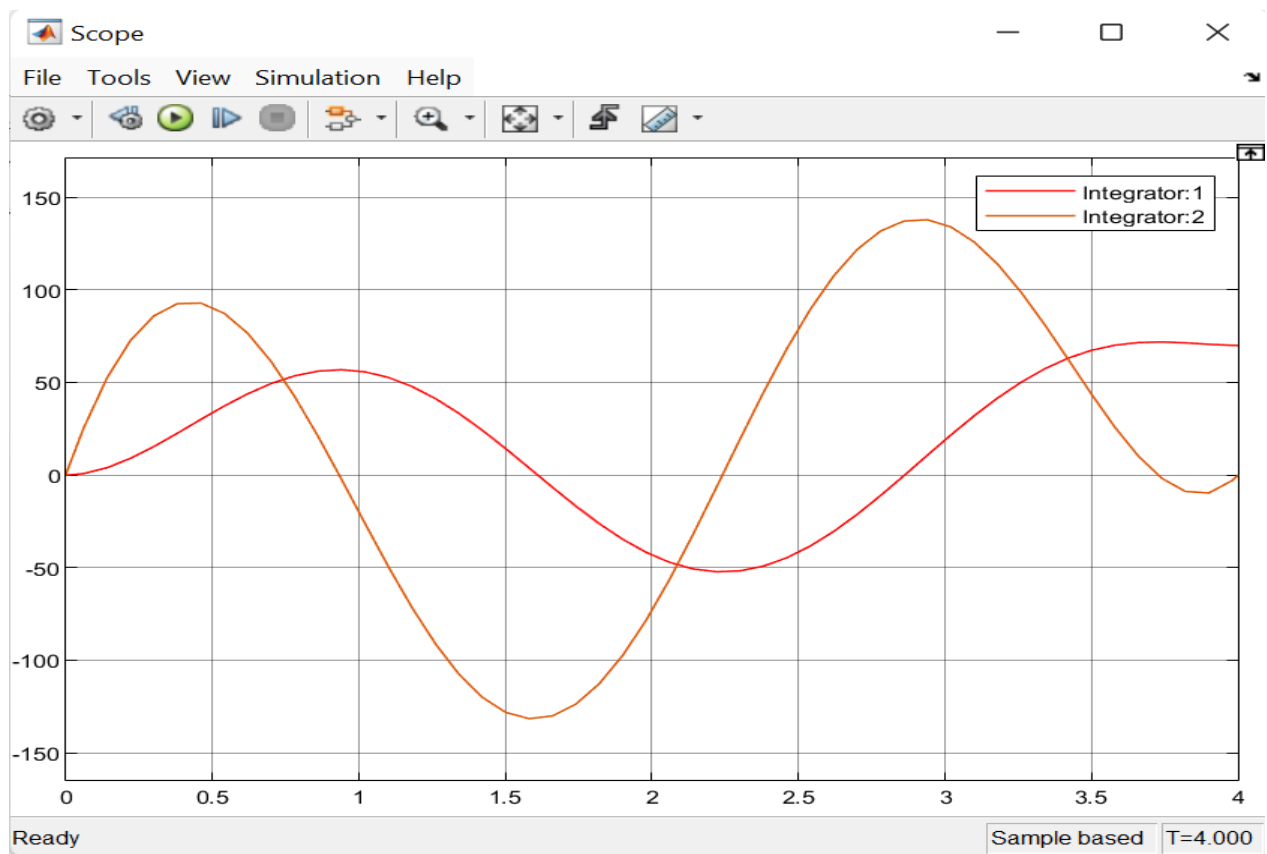


Figure 3. States x1 and x2

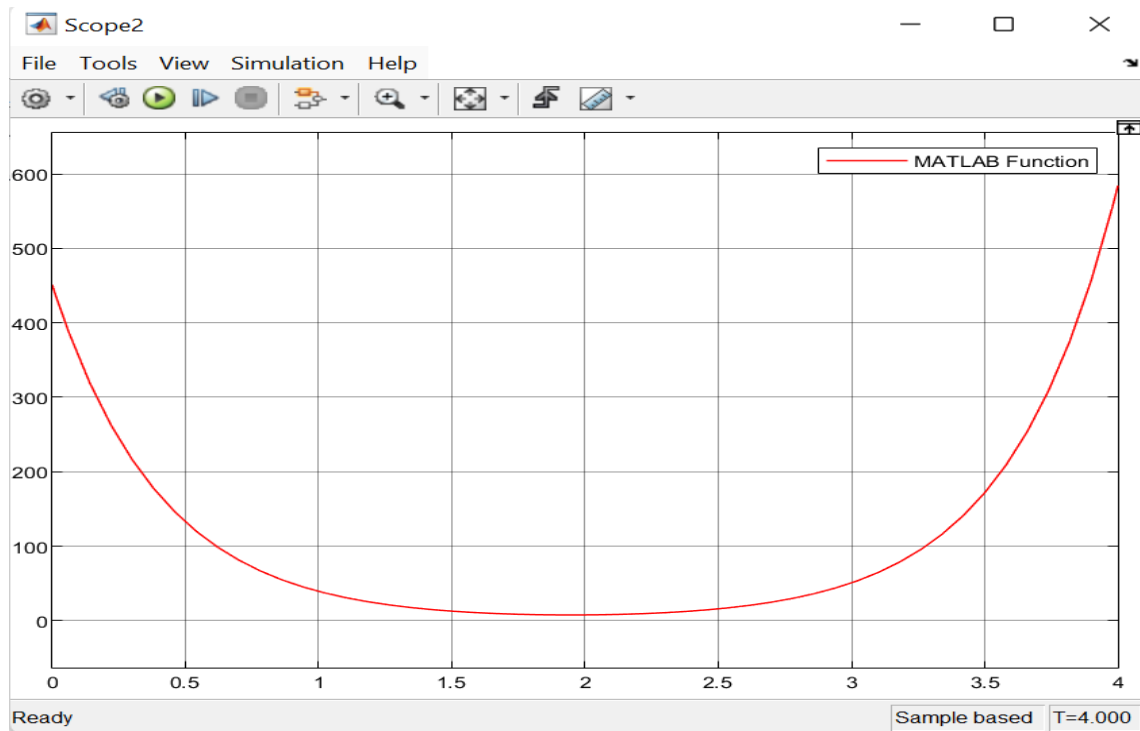


Figure 4. Control output

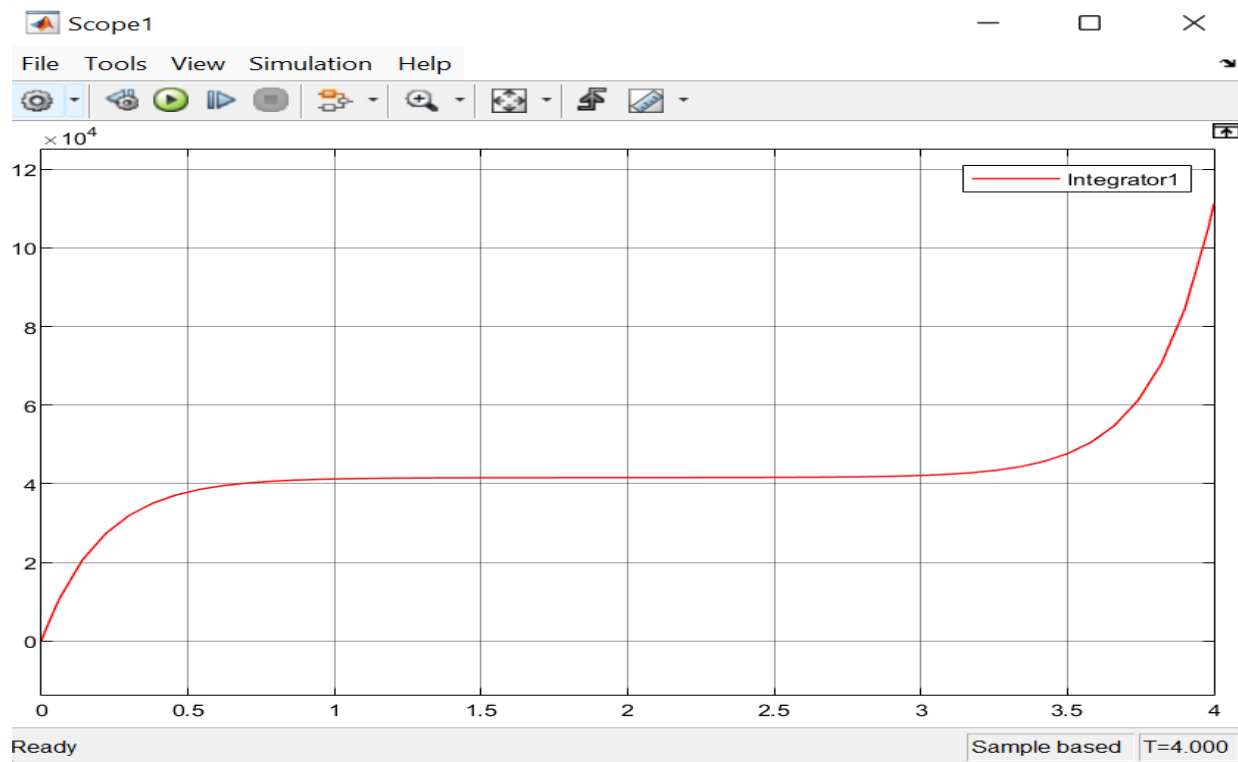


Figure 5. Cost function

We can observe that the controller is producing the correct output as the states are converging to the final states, now we will change the controller parameters to check if the cost function increases or not,

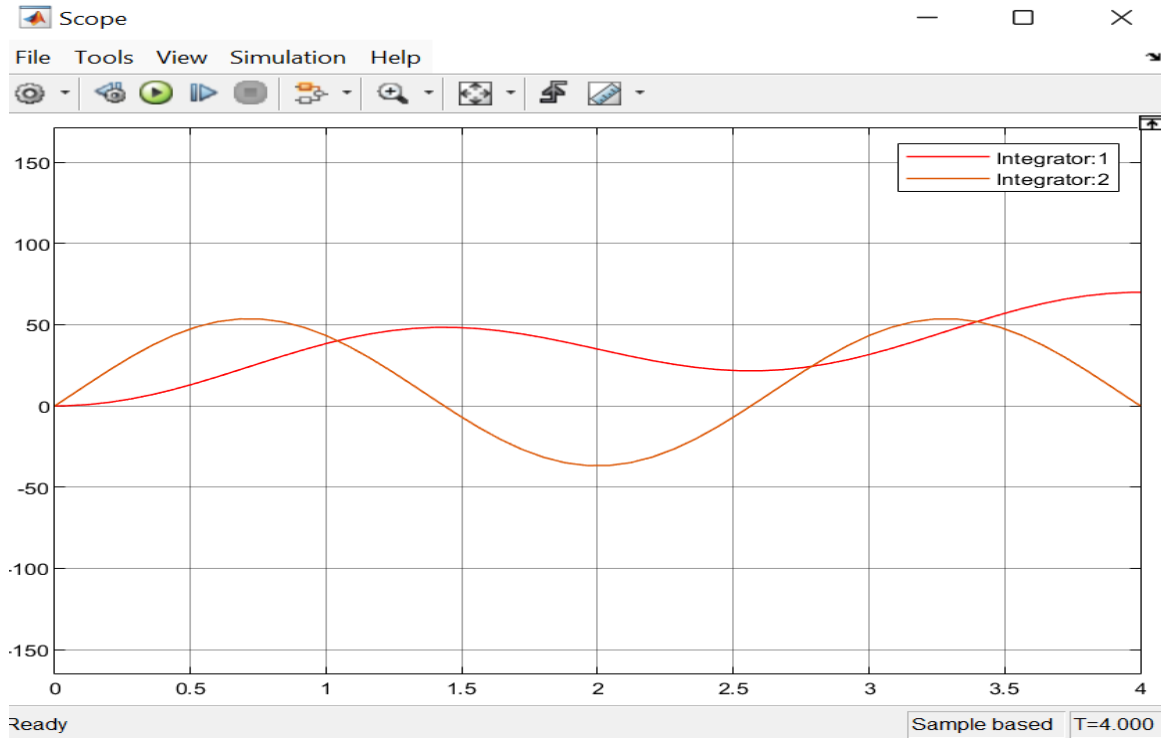


Figure 6. States x_1 and x_2 (Changed parameters)

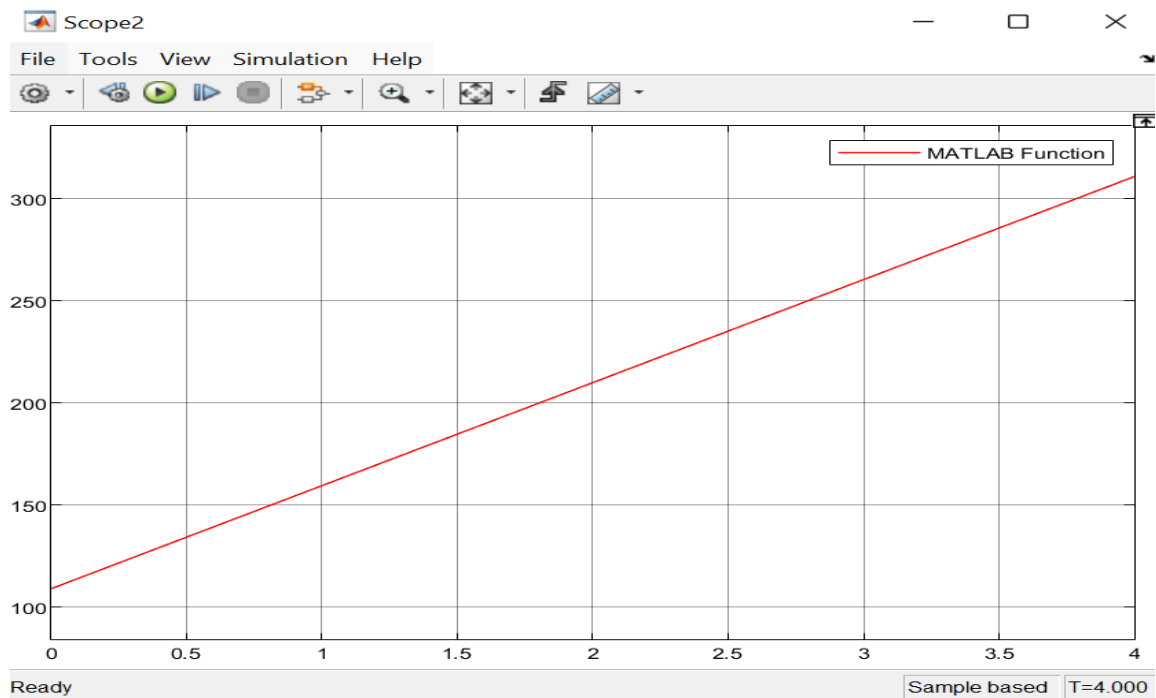


Figure 7. Control output (Changed Parameters)

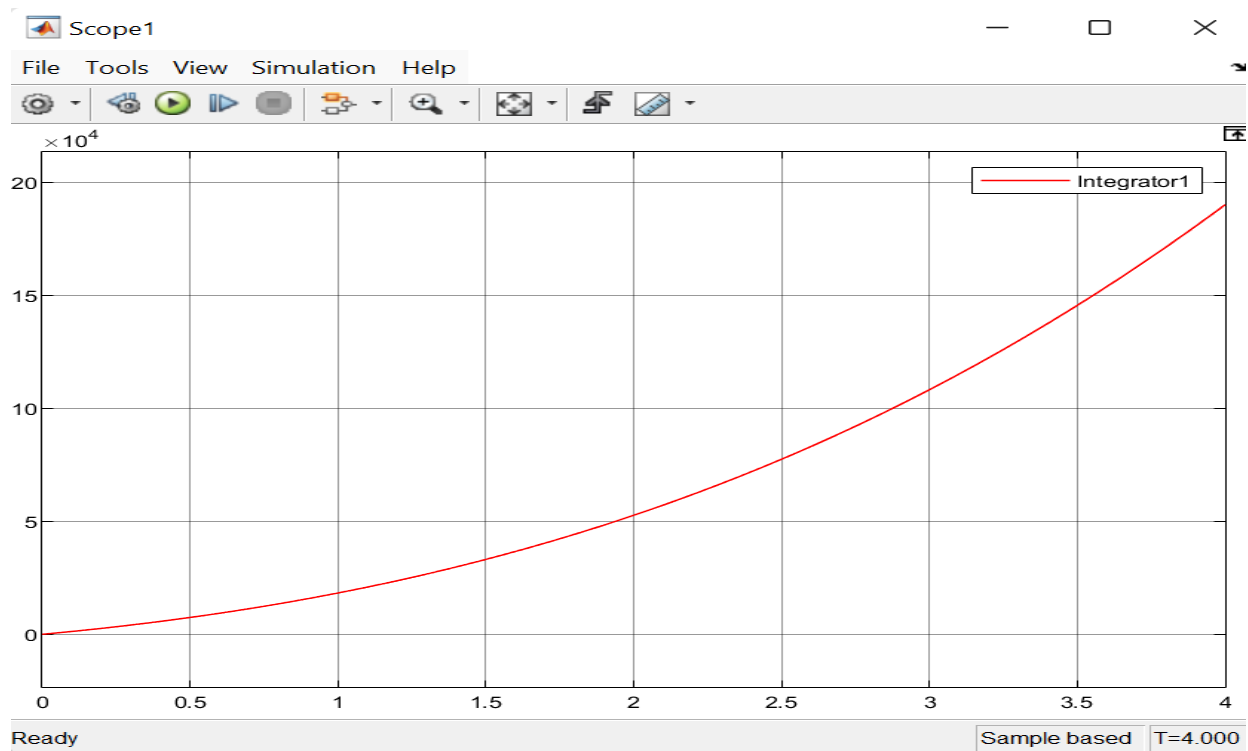


Figure 8. Cost function (Changed parameters)

Conclusion

In this laboratory work we have synthesized the optimal controller for the given plant and the cost function using the Hamiltonian and the euler lagrange equations by applying the pontryagins maximum principle,

We can conclude that the controller that we synthesized is an optimal controller and produces the required output with the minimum cost, as we also experimented by changing the controller parameters that the system is producing the same output but at a much largr cost since the cost function is increasing than the optimal one.